

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING – Combined Questions (CIA I & II)

2 Marks Questions

1. List the main steps in machine learning process.
2. What is cross-validation in machine learning?
3. Define activation function.
4. What is forward propagation?
5. What is the purpose of weights and biases in neural networks?
6. What are characteristics of a well-defined problem in AI?
7. What is the primary objective of Means-End Analysis?
8. Explain Instance and ISA relationships in semantic networks.
9. Give an example of frame representation.
10. What is a unit clause in resolution?

10 Marks Questions

1. Describe supervised learning with examples OR explain gradient descent.
2. Explain forward propagation with mathematical representation OR explain weight initialization impact.
3. Explain core components of state space problem OR discuss frame problem.
4. Discuss procedural vs declarative knowledge OR explain Generate and Test heuristic.

15 Marks Questions

1. Explain multilayer neural network structure.
2. Explain deep neural network architecture.
3. Explain working of Tanh activation function.
4. Compare forward and backward reasoning.
5. Explain hill climbing algorithm.
6. Discuss computable functions and predicates.